

## Unit 14, Lesson 5: Writing and Solving One-Step Addition and Subtraction Equations Within Real-World Contexts



### Benchmark

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.



### Learning Target

- ☐ I can write and solve one-step equations in one variable within a real-world context using addition and subtraction.



### 14.5.1 Warm-Up: Bell Work

1. Before last night, Faith had  $p$  Instagram posts on her account. She made 4 more posts last night. She now has 313 Instagram posts on her account. Write an equation that can be used to find  $p$ , the number of posts Faith had on her account before last night.



### 14.5.2 Collaboration: Solving Equations in Real-World Contexts

Work with your partner to complete the following.

1. Kiran's cat weighs 5 pounds less than Clare's puppy. Clare's puppy weighs 14 pounds.
  - a. Select all of the equations that could be used to find the weight of Kiran's cat.

- ☐  $x + 5 = 14$
    - ☐  $x - 5 = 14$
    - ☐  $x = 14 + 5$
    - ☐  $x = 14 - 5$
    - ☐  $x - 14 = 5$
    - ☐  $x + 14 = 5$
  - b. What does the variable,  $x$ , represent in this situation?
  - c. What is the weight of Kiran's cat?

2. Kiegan collected 7 more items than his cousin Joel did for the annual toy drive. Joel collected 12 items.
  - a. Select all of the equations that could be used to find the number of items Kiegan collected.

- ☐  $t = 7 - 12$
    - ☐  $t - 7 = 12$
    - ☐  $t + 7 = 12$
    - ☐  $t = 12 + 7$
    - ☐  $t + 12 = 7$
    - ☐  $t - 12 = 7$
  - b. What does the variable,  $t$ , represent in this situation?
  - c. How many items did Kiegan collect for the toy drive?

Discuss with your partner what you notice about the equations that represent the two situations. Summarize your discussion.



### 14.5.3 Guided Practice: Writing and Solving Real-World Equations to Find a Value

1. Jamal, a student at McLaughlin Middle School in Polk County, is preparing for a trip to Disney World with his mom and dad. An adult ticket is \$139. A child's ticket is \$5 less than an adult ticket. Jamal's parents are going to spend a total of \$412, before tax, on tickets.
  - a. Write an algebraic equation to represent the relationship between the cost of a child's ticket and the cost of an adult ticket in this situation.
  - b. What does the variable in your equation represent?
  - c. Solve the equation.
  - d. What was the cost of Jamal's ticket?
2. After doing some additional research, Jamal's mom also decided to purchase the meal bundle for their trip. The meal plan price is different for adults than children. It costs \$110 for his mom and dad. The total for all three meal plans was \$136. An equation can be written to find the unknown cost of Jamal's meal plan.
  - a. Write an algebraic equation that can be used to find the cost of Jamal's meal plan.
  - b. What does the variable in your equation represent?
  - c. Solve the equation.
  - d. How much was Jamal's meal plan?

3. Use the equation  $49 + t = 107$  to complete the following.

- Write a word problem that can be represented by the equation  $49 + t = 107$ .
- What does the variable represent based on the word problem you created?
- What is the value of the variable?
- Share **only** your word problem with your partner. Use your partner's word problem to fill in the table.

| Partner's Word Problem | What Does the Variable Represent? | Solve the Equation | Partner's Initials |
|------------------------|-----------------------------------|--------------------|--------------------|
|                        |                                   |                    |                    |



#### 14.5.4 Collaboration: Writing and Solving Equations for Real-World Contexts

Work with your partner to complete the following.

- PJ and Jasmine were given the following word problem.

*Lexi's school is having a food drive. At the end of today's lunch period, Lexi collected a total of 34 non-perishable items for the food drive. Lexi celebrated, as it was 11 more than yesterday's lunch collection! How many total non-perishable items were collected yesterday?*

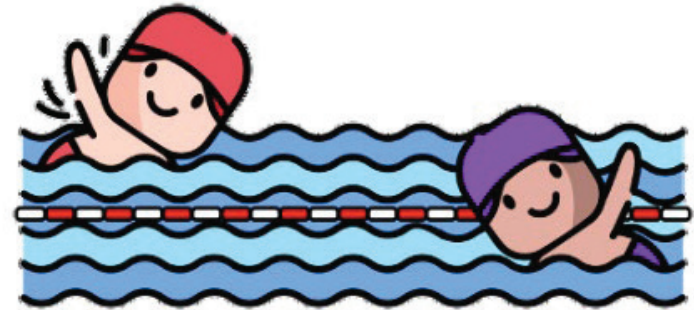
Jasmine told PJ they needed to solve the problem using an addition equation. PJ disagreed and said they need to use a subtraction equation.

- With your partner, determine who will follow Jasmine's suggestion and who will follow PJ's. Then, write the equation based on whose suggestion you are assigned and solve.

| Jasmine's Suggestion "Addition" | PJ's Suggestion "Subtraction" |
|---------------------------------|-------------------------------|
|                                 |                               |

- b. Compare your equations and discuss your solutions with your partner.
- c. Explain who was correct in their initial suggestion – Jasmine, PJ, both of them, or neither of them.
- d. How many non-perishable food items were collected during Lexi's lunch period yesterday?

2. An image of two swimmers in a pool is shown.



- a. Work with your partner to create an addition or subtraction word problem based on the image.
- b. Write an equation to represent your scenario.
- c. What does the variable represent in your equation represent?
- d. What is the value of the variable?



### 14.5.5 Your Turn!

Write and solve an algebraic equation for each of the following.

1. Priya and Gerri are working on an extra-credit poster for math class. Priya has 12 double-tip markers, which is 3 fewer than Gerri has. How many markers does Gerri have?
  
  
  
  
  
  
  
  
  
  
2. The most popular elective at Arts Academy is the improv class. Out of the 97 students that have applied, only 45 can be accepted. How many students are not accepted?



### 14.5.6 Wrap-Up: Reflection

1. Shade in the arrow up to the statement that best describes your current understanding of the benchmark below:

Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.



I'm so confident, I could explain how to write and solve one-step equations in one variable using addition or subtraction to someone else!

I can get to the right answer, but I don't understand well enough to explain how to write and solve one-step equations in one variable yet.

I understand some of this, but I don't understand all of it yet.

I have been trying hard, but I am finding this challenging.